SCHOOL
SUPERIOR
AVERAGE
ARTS
AND DESIGN
POLYTECHNIC
FROM PORTO

ALP

WORKSHEET
TEACHING ACTIVITY

Algorithmics and Data Structures

COURSE UNIT

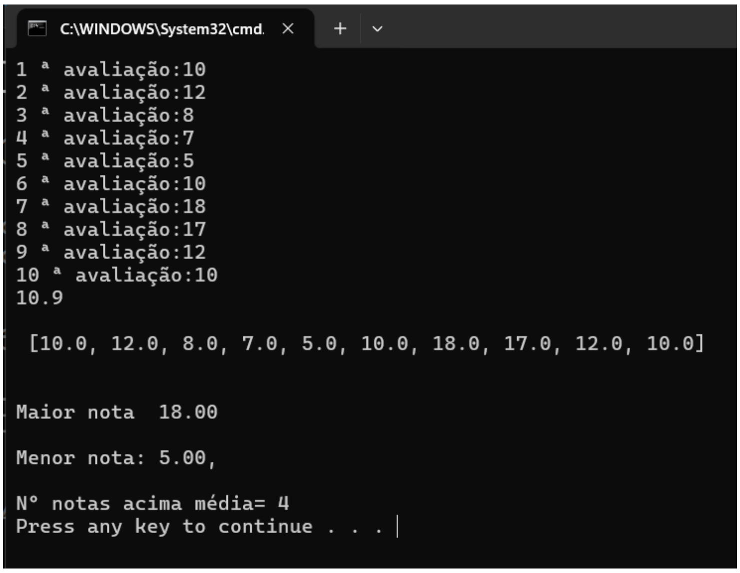
Sheet 04 – Lists

1. Implement a program to help a teacher to analyze the grades of their students. Program requirements:

   - Read student grades from user input and store them in a **list.** The student's grades must be between [0-20] in float format (input validated with exceptions). The user should enter the grades of 10 students.
   - After all grades have been entered, the program should call the function named **data-Grades** (*list*). The function receives the list of grades and should return:

     The **highest** and **lowest** grades.

     How many grades are **above average?**



2. Create a **generateEuromillionKeys()** function that allows you to generate Euromillions keys randomly (5 integers between 1 and 50, as well as the stars: two stars between 1 and 12).

- Note that the numbers and stars in a key cannot be repeated.
- The function should return a key ordered (5 numbers and 2 stars)
- At the end, print the generated key and ask the user if he wants to generate a new key (S/N).

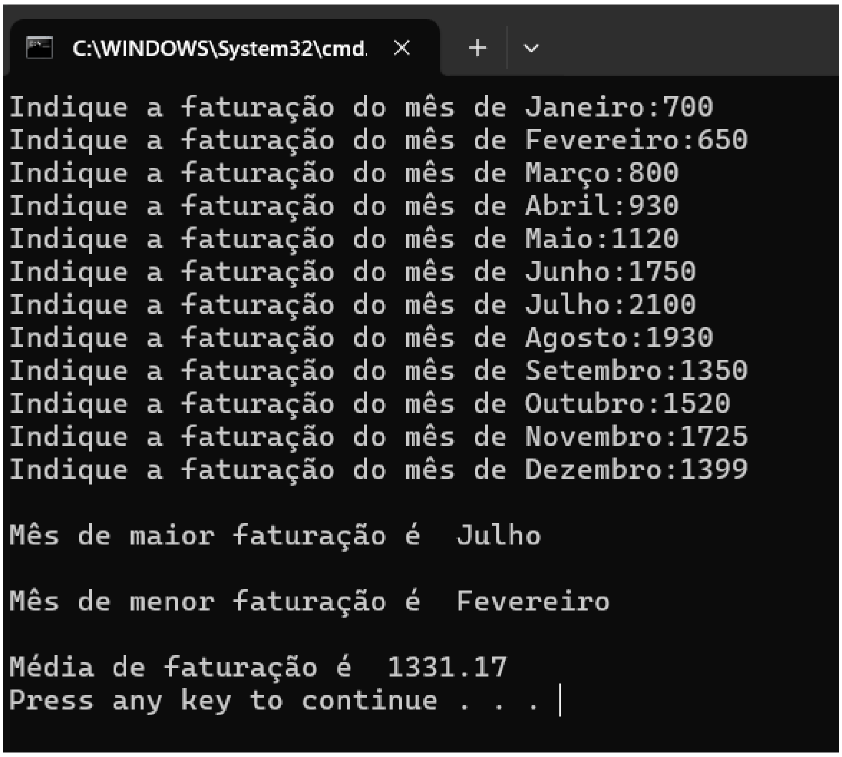
```
C:\WINDOWS\System32\cmd.

 Chave do Euromilhões: [14, 26, 28, 36, 44]      Estrelas: [7, 10]
Deseja gerar nova chave(S/N)?:s
 Chave do Euromilhões: [2, 4, 14, 37, 39]        Estrelas: [1, 5]
Deseja gerar nova chave(S/N)?:
```

3. Create a program that allows you to generate a random number **between [1-100]** and display it in the Console. Then your program should ask the user: "*Do you want to generate a new number (S/N)?*".
   - If you answer **No**: end the program, displaying the list of all the numbers generated, in descending order;
   - If you answer **Yes**: you must generate a new random number **between the last generated number +1, and 100**

   Notes:
   - It must be ensured that the random numbers generated in the previous process **are never** repeated;
   - If the randomly generated number is equal to 100, you must finish the program by displaying the list of all the generated numbers, in descending order

```
                         Numero sorteado:  8
Quer gerar um novo número? (S/N)? s

                         Numero sorteado:  42
Quer gerar um novo número? (S/N)? s

                         Numero sorteado:  96
Quer gerar um novo número? (S/N)? s

                         Numero sorteado:  98
Quer gerar um novo número? (S/N)? s

                         Numero sorteado:  100
[100, 98, 96, 42, 8]
Press any key to continue . . .
```

4. Develop a program that allows you to input a company's monthly sales over the 12 months of the year (from January to December). After that, the program should include the call of 3 functions which they <u>return</u>, respectively:
    - the **month** with the highest value of sales;
    - the **month** with the lowest value of sales;
    - The **average** monthly sales.



5. Develop a program that reads a list of 10 integers (validate with exceptions). The list may contain duplicate values.
   Then, given a certain search value (by input),call a function named **searchNumber** *()* that should <u>return **the positions**</u> where it finds the search value in the list.
   If the search value does not exist in the list, a suitable message should appear.

```
C:\WINDOWS\System32\cmd.
1 ° Número : 10
2 ° Número : 12
3 ° Número : 14
4 ° Número : 10
5 ° Número : 18
6 ° Número : 10
7 ° Número : 8
8 ° Número : 8
9 ° Número : 19
10 ° Número : 15
Valor de pesquisa:10
 O numero 10  existe na lista na(s) posição(ões)  1 4 6
Press any key to continue . . .
```

6. Implement a program that allows you to read the number of visitors to an exhibition, which runs from Sunday to Saturday.

   Next, create a function that allows you to list the number of daily visitors in descending order, as shown in the image below.

   Also indicate, at the end, the average number of visitors per day (with 2 decimal places) and the day that came closest to the average number of visitors.

```
C:\WINDOWS\System32\cmd.
N° visitantes na/no Domingo     : 100
N° visitantes na/no Segunda     : 45
N° visitantes na/no Terça       : 50
N° visitantes na/no Quarta      : 70
N° visitantes na/no Quinta      : 62
N° visitantes na/no Sexta       : 81
N° visitantes na/no Sábado      : 120
Sábado           120
Domingo          100
Sexta            81
Quarta           70
Quinta           62
Terça            50
Segunda          45
N° médio de visitantes: 75.43
Dia mais próximo do valor médio: Quarta
```

7. Create a function that receives a text (text can be entered by the user) and a search word. Your function should indicate how many times that word occurs in the text.

   Note:

   - Consider that the search word is not case sensitive, that is, it does not differentiate between uppercase and lowercase letters.

8. Develop a program that allows you to manage a list / queue (such as in a supermarket, or in customer service) with a maximum capacity of 20 tickets.

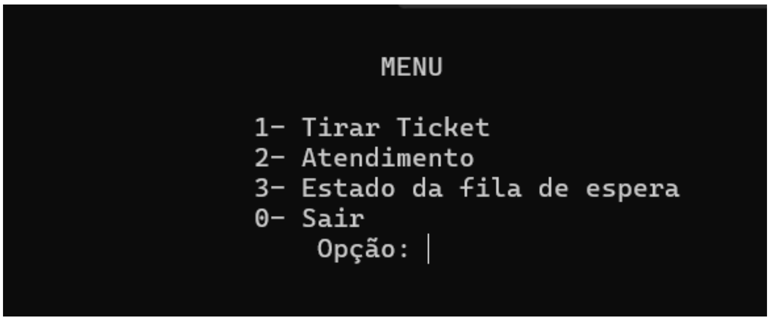
   When the program starts, the first ticket available is the 1.

   Queue layout (list with 20 positions):

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

   Your program should contain a menu with the following options:

MENU
1 – Get a ticket
2 – Call a ticket
3 - List status
0 - Exit



- **Get a ticket**

  You must call a function that simulates what ticket number is available in the list/queue.

  If the waiting list queue has 20 tickets, a message should appear, such as: "It is not possible to generate more tickets at the moment".

- **Call a ticket**

  You must call a function that allows you to always attend to the element that is in the first position of the queue at a given time (you should print your ticket number on the console).

  In this case, all other members of the queue must move forward to one position.

- **List status**

  You must call a function that prints the number of tickets that are waiting and the number of available tickets in the queue.

- Exit

  Program execution ends